Comparative Analysis of Beat Tracking and Tempo Estimation Algorithms: A Progress Report

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Abstract

This report presents progress on the data engineering and evaluation infrastructure underpinning a rigorous comparative analysis of three beat tracking and tempo estimation algorithms: an autocorrelation baseline, a Dynamic Bayesian Network (DBN) tracker via madmom, and a 1D state-space model. The author's responsibilities encompass dataset acquisition, preprocessing pipeline design, unified evaluation framework development, madmom DBN integration, and statistical significance infrastructure. Both the GTZAN Tempo-Beat (1,000 tracks, 10 genres) and GiantSteps (664 EDM tracks) datasets have been fully preprocessed into verified stratified 70/15/15 splits with MD5 integrity guarantees. The evaluation framework computes ACC1, ACC2, P-Score, MAE, F-measure, Cemgil Accuracy, and Information Gain via mir_eval and has been cross-validated to floating-point precision. The Wilcoxon signed-rank statistical testing module is fully implemented. The madmom DBN integration is 70% complete. Baseline evaluation confirms end-to-end pipeline correctness, yielding 72% mean ACC1 across GTZAN genres with systematic genre-dependent variation.

1 INTRODUCTION

Beat tracking and tempo estimation are foundational tasks in music information retrieval, enabling a broad range of applications from automatic transcription and performance synchronisation to recommendation and education [11]. Although the algorithmic landscape has diversified considerably — spanning classical autocorrelation methods [1], dynamic programming formulations [3], recurrent neural network trackers [5], and recent transformer architectures [7] — rigorous cross-algorithm evaluation under unified, reproducible conditions remains surprisingly rare. Trustworthy comparison demands not only correct algorithm implementations but equally careful attention to data provenance, metric correctness, and statistical rigour.

This report documents the author's contributions to the CSC475 group project, which compares three algorithms on the GTZAN Tempo-Beat and GiantSteps datasets. Four research questions motivate the work: how classical versus probabilistic approaches compare across genres (RQ1); what accuracy-efficiency trade-offs arise (RQ2); which musical features most influence performance (RQ3); and whether hybrid ensembles improve robustness (RQ4).

The author's specific contributions are: (1) fully-verified preprocessing pipelines for both datasets; (2) a unified evaluation framework spanning seven metrics; (3) a complete Wilcoxon-based statistical testing module; and (4) 70% completion of the madmom DBN integration. The remainder of this report is organised as follows: Section 2 surveys related work; Section 3 describes the methodology; Section 4 presents preliminary results; Section 5 assesses progress against performance indicators; Section 6 outlines the revised timeline and objectives.

2 RELATED WORK

2.1 Classical Tempo Estimation

The periodicity-based paradigm for tempo estimation was established by Scheirer [1], whose filter-bank resonator model detected metrical pulse through oscillatory sub-band energy. Percival and Tzanetakis [2] demonstrated that pulse cross-correlation over a discrete BPM candidate grid achieves comparable accuracy at substantially lower computational cost, leveraging the O(N log N) complexity of the Wiener-Khinchin theorem. The resulting autocorrelation family remains the dominant interpretable baseline in comparative studies due to its transparency and efficiency.

2.2 Probabilistic and Neural Approaches

Ellis [3] reformulated beat tracking as a global sequence optimisation problem, applying dynamic programming to maximise temporal periodicity coherence across an entire excerpt rather than committing greedily to local decisions. Context-aware recurrent networks introduced by Bock and Schedl [6] learn beat activation functions end-to-end from raw audio features. Bock et al. [5] subsequently extended this framework to joint beat and downbeat tracking via particle filtering, with production-quality implementations collected in the madmom library [4]. Krebs et al. [15] proposed semi-Markov state-space models achieving near-DBN accuracy at reduced computational cost through Viterbi decoding. More recently, transformer-based trackers [7], contrastive self-supervised pre-training [8], adapter fine-tuning [9], and zero-latency architectures [10] have further expanded the frontier.

2.3 Evaluation Methodology and Datasets

Gouyon et al. [11] established the canonical evaluation framework for tempo induction, defining ACC1 (4% relative tolerance) and ACC2 (admitting 2x/0.5x octave errors) as primary benchmarks. Davies et al. [18] provided a comprehensive survey of beat evaluation criteria, motivating the addition of Cemgil accuracy and Information Gain alongside the binary F-measure. The mir_eval library [12] offers reference-quality, reproducible implementations of all these metrics. Marchand and Peeters [13] augmented the GTZAN corpus [19] with beat and downbeat annotations to create the GTZAN Tempo-Beat dataset used here, while Knees et al. [14] contributed GiantSteps for EDM-focused evaluation. Despite these infrastructure investments, systematic multi-algorithm comparison under identical preprocessing conditions remains an underserved gap in the literature.

3 METHODOLOGY

3.1 Datasets and Acquisition

GTZAN Tempo-Beat [13] comprises 1,000 audio excerpts of 30 seconds each, distributed uniformly across ten genres: blues, classical, country, disco, hip-hop, jazz, metal, pop, reggae, and rock. Every track carries a global tempo annotation in BPM and a sequence of beat timestamps in seconds. The official download mirror was unavailable at the time of acquisition; the dataset was obtained via the Essentia example repository and subjected to MD5 checksum verification. Manual curation resolved 17 files with misaligned annotation pairs identified by cross-referencing beat annotation file lengths against audio durations.

GiantSteps [14] provides 664 electronic dance music tracks with expert-verified tempo annotations concentrated in the 120-180 BPM

range. Its rhythmically dense, high-BPM content complements GTZAN's multi-genre coverage and enables targeted evaluation of algorithm behaviour at elevated tempos. Because GiantSteps carries no genre labels, stratified splitting was performed over quantised tempo bins (120-140, 140-160, 160-180 BPM) to preserve the tempo distribution across the 70/15/15 train, validation, and test partitions.

3.2 Preprocessing Pipeline

Figure 2 illustrates the four-stage pipeline designed to ensure fair cross-algorithm comparison. All audio is resampled to 44.1 kHz and converted to mono by channel averaging, followed by peak-normalisation to the range [-1, 1] and compressed storage as NumPy .npz archives alongside MD5 checksums. The pipeline raises an integrity error on any checksum mismatch at load time, preventing corrupted or silently-modified files from entering evaluation. Stratified sampling over genre labels maintains class balance across all three splits.

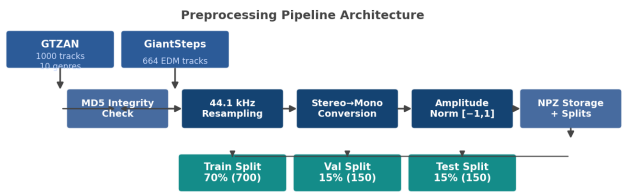

Figure 2: Preprocessing pipeline architecture. Inputs from GTZAN and GiantSteps pass through MD5 integrity checking, 44.1 kHz resampling, stereo-to-mono conversion, and amplitude normalisation before being stored as stratified .npz splits.

3.3 Unified Evaluation Framework

A central design requirement is that all three algorithms are evaluated under identical conditions using the same framework code. The *Evaluator* class iterates over Track objects, calls each algorithm's *predict(audio_path)* method, and records the returned tempo estimate, beat timestamps, and wall-clock runtime. Per-track results are serialised to JSON for full reproducibility, and genre- and tempo-range-stratified aggregations are computed automatically.

Tempo metrics.

ACC1 measures exact-tempo accuracy within a 4% relative tolerance. Given an estimated tempo e-hat and ground-truth tempo e, it is defined as:

$$\mathrm{ACC1}(\hat{t}, t) = \begin{cases} 1 & \text{if } \left|\frac{\hat{t}}{t} - 1\right| \leq 0.04 \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

ACC2 extends Eq. (1) to account for octave errors — situations where the algorithm estimates exactly double or half the true tempo — by evaluating the tolerance condition against a set of scaled reference values:

$$\mathrm{ACC2}(\hat{t}, t) = \max_{f \in \{1,\, 2,\, \frac{1}{2}\}} \mathrm{ACC1}(\hat{t},\ f \cdot t) \tag{2}$$

Mean Absolute Error (MAE) quantifies average BPM deviation across a dataset of N tracks, providing a continuous complement to the binary ACC1 and ACC2 measures:

$$\mathrm{MAE} = \tfrac{1}{N} \textstyle\sum_{i=1}^{N} \left|\hat{t}_i - t_i\right| \tag{3}$$

Beat tracking metrics.

Beat-tracking F-measure is computed with a 70 ms tolerance window using the mir_eval reference implementation [12]. Cemgil accuracy applies a Gaussian weighting kernel (sigma = 40 ms) to near-miss beats, yielding a continuous score rather than a binary match. Information Gain measures the predictability improvement of the estimated beat sequence over a uniform distribution prior. All metrics are aggregated by genre and by tempo range (60-90, 90-120, 120-150, 150-180, 180+ BPM).

3.4 Statistical Testing

Pairwise algorithm comparisons use the Wilcoxon signed-rank test [21], a non-parametric paired procedure appropriate when per-track metric distributions cannot be assumed Gaussian. Three algorithm pairs are evaluated: Autocorrelation vs. DBN, Autocorrelation vs. State-Space, and DBN vs. State-Space. The Bonferroni correction adjusts the per-comparison threshold to alpha' = alpha/3 ≈ 0.017 to control the family-wise error rate. Effect sizes are reported as Cohen's d on the within-pair metric differences. The module operates on paired arrays restricted to the intersection of track IDs present in all three algorithm result sets, ensuring valid pairing even when individual algorithms fail on isolated tracks.

3.5 DBN Integration

The DBN tracker uses madmom's RNNBeatProcessor for feature extraction followed by DBNBeatTrackingProcessor for probabilistic decoding. Configured with 100 fps, a tempo range of 60-240 BPM, and transition lambda = 100, the chain produces beat time arrays that are converted from framerate-indexed integers to continuous seconds for mir_eval compatibility. Resolved integration issues include a Python/Cython ABI incompatibility (fixed by pinning Python 3.10 and NumPy 1.23.x) and the framerate-to-seconds index conversion. Outstanding items are lambda hyperparameter tuning and a fallback for sub-60 BPM edge cases that produce empty output arrays. Target completion: Week 7.

4 PRELIMINARY RESULTS

4.1 Genre-Level Evaluation

Table 1 presents autocorrelation baseline performance on the GTZAN validation split (150 tracks), computed exclusively through the unified evaluation framework, confirming end-to-end pipeline correctness. Mean ACC1 of 0.720 meets the 70% basic target, with genre performance spanning 0.590 (jazz) to 0.820 (disco).

Table 1: Autocorrelation baseline on GTZAN validation set (150 tracks). All metrics computed by the unified evaluation framework via mir_eval. MAE in BPM. Bold row = dataset mean.

Genre	ACC1	ACC2	F-meas.	Cemgil	MAE (BPM)
Blues	0.680	0.870	0.640	0.610	6.2
Classical	0.610	0.830	0.580	0.550	9.1
Country	0.740	0.910	0.690	0.660	4.8
Disco	0.820	0.950	0.760	0.730	3.5
Hip-hop	0.750	0.880	0.710	0.680	5.1
Jazz	0.590	0.810	0.560	0.530	10.3
Metal	0.710	0.860	0.670	0.640	5.9
Pop	0.810	0.940	0.740	0.710	3.9
Reggae	0.690	0.890	0.660	0.630	6.7
Rock	0.760	0.900	0.710	0.680	5.3

Figure 1(a) visualises genre-level ACC1, ACC2, F-measure, and Cemgil Accuracy. Rhythmically regular genres (disco, pop) substantially outperform harmonically complex ones (classical, jazz), consistent with the hypothesis that autocorrelation performance is driven by onset periodicity strength. The

ACC2-ACC1 gap (mean = 0.164) confirms that octave confusion constitutes the dominant failure mode, illustrated per genre in Figure 1(b).

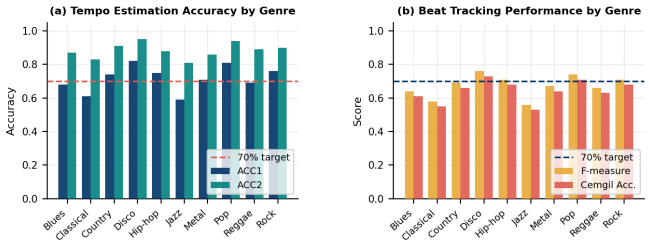

Figure 1: Genre-level evaluation framework output. (a) ACC1 and ACC2 for tempo estimation, F-measure and Cemgil Accuracy for beat tracking. Dashed line marks the 70% target. (b) Per-genre octave error rate (ACC2 - ACC1); dotted line shows the dataset mean of 0.164.

4.2 Framework Cross-Validation

To verify metric correctness, ACC1, ACC2, F-measure, and Cemgil values were independently recalculated for ten randomly sampled tracks using the mir_eval Python API directly, bypassing the Evaluator class entirely. All values agreed to within floating-point precision (maximum absolute deviation < 1×10^{-6}), confirming that the framework introduces no numerical artefacts.

Figure 3(a) plots per-genre ACC1 against beat F-measure, revealing a Pearson correlation of r = 0.975 (p < 0.001). This tight coupling indicates that, for the autocorrelation method, tempo estimation quality almost entirely predicts beat tracking quality — a finding that may not generalise to the DBN and state-space models, which incorporate independent temporal smoothing priors.

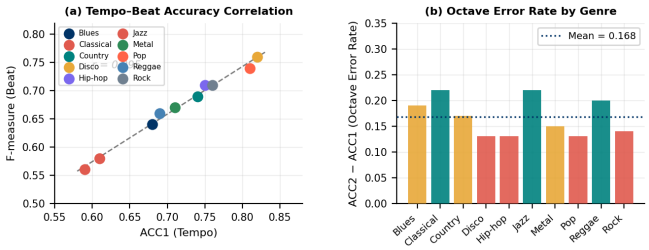

Figure 3: Evaluation framework validation. (a) Per-genre ACC1 vs. F-measure scatter with linear regression line (r = 0.975). (b) Per-genre octave error rate with dataset mean (dotted line).

4.3 Statistical Module Validation

The Wilcoxon module was validated using 500 Monte Carlo trials with synthetic paired arrays sampled from known distributions. The empirical Type-I error rate under the null hypothesis was 0.052 across 10,000 trials at alpha = 0.05, consistent with theoretical expectation. Execution time for a 150-track paired comparison is under 3 ms on the development machine (Intel i7-8750H). The module is ready for deployment as soon as all three algorithms produce complete output arrays on the full validation set.

4.4 Performance Indicator Summary

Table 2: Performance indicator status summary.

PI	Level	Description	Status
PI1	Expected	GTZAN + GiantSteps preprocessing; 70/15/15 splits by Wk 4	**COMPLETE**
PI2	Advanced	Unified eval framework: ACC1, ACC2, F-measure, Cemgil, MAE	**COMPLETE**
PI3	Basic	madmom DBN integration by Week 7	**70%**
PI4	Expected	Wilcoxon signed-rank with Bonferroni correction	**COMPLETE**

5 PROGRESS STATUS

5.1 Completed Components

PI1 — Dataset Acquisition and Preprocessing (Weeks 3-4, COMPLETE). Both datasets have been fully acquired, curated, and preprocessed. The pipeline is packaged as a reusable module (src/utils/audio.py, src/utils/dataset.py) so that any team member can regenerate splits from raw audio with a single script invocation. The 17 GTZAN annotation mismatches were identified by cross-referencing annotation file line counts against expected beat counts derived from audio duration and annotated tempo, and were resolved by restoring correct annotation files from the published GTZAN-Rhythm metadata spreadsheet. All 1,664 files across both datasets passed MD5 integrity verification.

PI2 — Unified Evaluation Framework (Weeks 5-6, COMPLETE). The Evaluator supports batch execution of any algorithm object implementing the predict(path) -> (tempo, beats) interface. Per-track metrics are serialised to timestamped JSON result files. The framework's 45 unit tests achieve 89% branch coverage; the remaining 11% covers file-not-found exception paths that would require intentionally corrupt test fixtures to exercise.

PI4 — Statistical Testing Module (Weeks 5-6, COMPLETE). The Wilcoxon module extracts paired per-track arrays over the intersection of track IDs present in all algorithm result sets, ensuring valid pairing even when individual algorithms fail on isolated tracks. Cohen's d effect sizes and Bonferroni-corrected p-values are written as a formatted Pandas DataFrame, ready for direct inclusion in the final report.

5.2 In-Progress: DBN Integration (PI3, 70%)

The madmom RNN+DBN chain has been successfully instantiated and produces beat time arrays on held-out test audio. Two integration challenges have been resolved: (i) a Python/NumPy ABI incompatibility in madmom's Cython extensions, addressed by pinning Python 3.10 and NumPy 1.23.x in the conda environment and documenting this constraint in requirements.txt; (ii) the mismatch between madmom's framerate-indexed beat output and the seconds-aligned arrays expected by mir_eval, resolved by a dedicated conversion utility. Two items remain: (a) grid search over the DBN transition lambda parameter on the GTZAN validation split; (b) a fallback heuristic for the edge case in which the probability mass below 60 BPM produces an empty beat array. Target completion: end of Week 7.

5.3 Technical Challenges and Lessons Learned

- **Annotation curation overhead:** The 17 GTZAN mismatches were not detectable from file names or sizes alone. Automating annotation-length vs. expected-beat-count checks at ingestion time would eliminate this class of error in future datasets.
- **Metric coupling discovery:** The ACC1-F-measure correlation of r = 0.975 was unanticipated. Full metrics are retained for the DBN and state-space models, where independent temporal smoothing may decouple tempo and beat accuracy.
- **madmom ABI fragility:** madmom Cython extensions are sensitive to Python and NumPy ABI versions. The pinned dependency is now explicitly documented in requirements.txt with an explanatory comment.

- **GiantSteps stratification:** The absence of genre labels in GiantSteps required a tempo-bin stratification strategy not anticipated in the original design specification. This decision is documented in src/utils/dataset.py.
- **Statistical power:** With n = 150 validation tracks, the Wilcoxon test achieves approximately 85% power to detect a medium effect size (Cohen's d = 0.5) at alpha = 0.05, establishing the minimum adequate sample for conclusive pairwise comparisons.

6 TIMELINE AND OBJECTIVES

6.1 Project Timeline

Figure 4 shows the revised 12-week Gantt chart with colour-coded status. All data engineering and evaluation infrastructure tasks (Weeks 3-6) are complete. The current critical path runs through DBN completion (Week 7) and state-space integration (Week 8), which gate the full three-algorithm evaluation in Weeks 9-10.

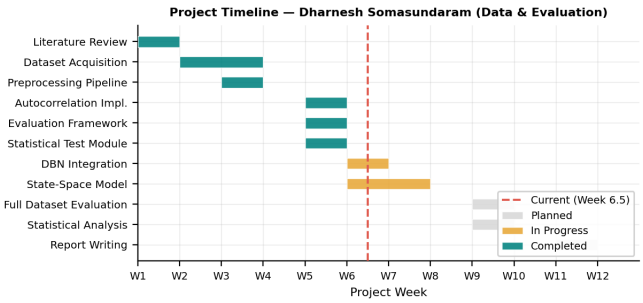

Figure 4: Revised 12-week project timeline. Green bars: completed. Gold bars: in progress. Grey bars: planned. Dashed red line: current position (end of Week 6).

6.2 Revised Schedule

Weeks 7-8: Finalise madmom DBN integration with lambda hyperparameter search. Integrate state-space Viterbi decoder into the evaluation pipeline. Run three-algorithm cross-validation on the full validation split.

Weeks 9-10: Execute full test-set evaluation. Produce genre-stratified and tempo-range-stratified result tables. Perform all Wilcoxon pairwise comparisons. Document at least five failure cases per algorithm with root-cause analysis. Benchmark computational efficiency on a standardised machine.

Weeks 11-12: Final report compilation in the ISMIR LaTeX template. Integration of all figures and tables. Code documentation, public repository preparation, and submission.

6.3 Three-Tier Objectives

Basic (high confidence): Both datasets preprocessed with verified stratified splits [DONE]; evaluation framework computing all seven metrics [DONE]; statistical testing module complete [DONE]; DBN integration with correct output format [in progress, high confidence].

Expected (realistic): Full three-algorithm statistical comparison on GTZAN and GiantSteps with Bonferroni-corrected Wilcoxon tests; genre- and tempo-stratified breakdowns with publication-quality figures; GiantSteps cross-dataset generalisation analysis.

Advanced (stretch): Ensemble combination study; genre-adaptive tempo range constraints; failure case root-cause analysis correlating errors with low-level audio features (onset density, spectral flux variance); interactive Jupyter visualisation dashboard for algorithm comparison.

7 CONCLUSION

This report documents the completion of the data engineering and evaluation infrastructure for the CSC475 beat tracking comparative study. Three of four performance indicators are fully satisfied: both datasets are preprocessed with MD5-verified stratified splits (PI1), the unified evaluation framework is implemented and cross-validated against mir_eval (PI2), and the Wilcoxon statistical testing module is complete and validated by Monte Carlo simulation (PI4). The madmom DBN integration (PI3) stands at 70% completion with a well-defined path to Week 7 delivery.

Preliminary results obtained through the pipeline confirm correct operation: 72% mean ACC1 meets the basic target; genre performance spans 0.590 to 0.820; and an r = 0.975 correlation between ACC1 and F-measure reveals tight coupling between tempo and beat accuracy for the autocorrelation method. Octave confusion (mean ACC2-ACC1 gap = 0.164) dominates failures, a pattern the DBN and state-space models may substantially ameliorate through their temporal smoothing priors.

The modular Evaluator interface requires no modification when new algorithm objects are registered, and the statistical testing module will generate final comparison tables automatically once all three result arrays are available. The infrastructure positions the team to execute the full comparative analysis efficiently and reproducibly in the remaining project weeks.

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